

ramanfit_swnt

June 30, 2026

1 SWNT Raman analysis

Single-wall carbon nanotube spectra, using the `ramanfit_swnt` package:

load → **Si calibration** → **RBM** (→ **diameter**, **candidate (n,m)**) → **joint D-G fit** → **2D** → **quality metrics**

Unlike the amorphous/MW-carbon D/G fit in `ramanfit.ipynb`, this:

- models the low-frequency **RBM** band (diameter fingerprint), one Lorentzian per detected sub-band, above the 180 cm^{-1} notch-filter edge,
- fits the whole **D-G region jointly over $1000\text{--}1700\text{ cm}^{-1}$** with five Lorentzians — **D**, a broad **D** disorder band ($\sim 1500\text{--}1535$), the **G /G doublet**, and **D** (~ 1600) — so the D-to-G valley is modelled instead of forced onto a straight baseline ($R^2 = 0.999$ vs 0.98 for isolated windows),
- fits the full **second-order region** (G^* , the 2D triplet, D+D, 2D, 2D) on a flat baseline,
- calibrates the shift axis against the **Si 520.7 cm^{-1}** substrate line.

```
[1]: import matplotlib.pyplot as plt
from ramanfit_swnt import (load_spectrum, analyze_swnt, SwntConfig,
                           choose_spectrum_file)
from ramanfit_swnt.report import plot_analysis, summary_text

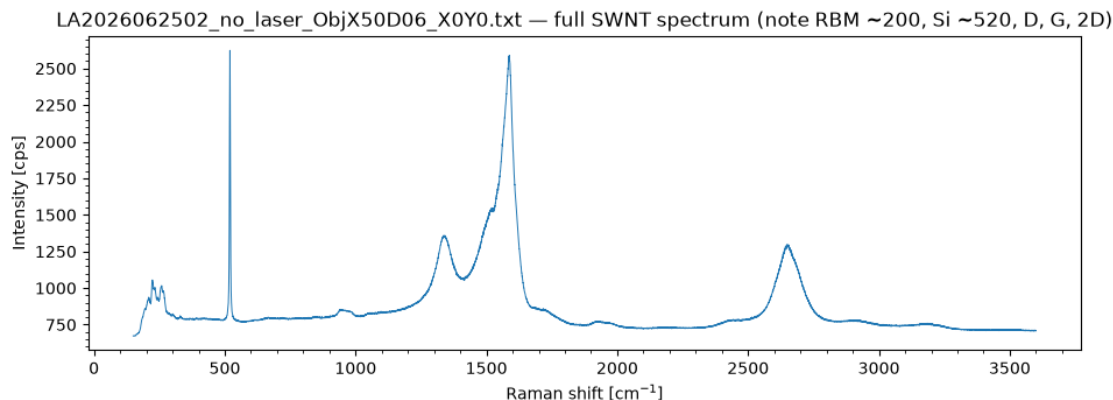
DEFAULT_INFILE = "data/LA2026062502_no_laser_ObjX50D06_X0Y0.txt" # 532 nm SWNT_
    ↪ on Si
# Pop up a file chooser; cancel (or a headless run) keeps the default.
INFILE = choose_spectrum_file(default=DEFAULT_INFILE) or DEFAULT_INFILE

x, y = load_spectrum(INFILE)
print(f"{INFILE}: {len(x)} points, {x[0]:.0f}-{x[-1]:.0f} cm-1")
```

data/LA2026062502_no_laser_ObjX50D06_X0Y0.txt: 25732 points, 150–3600 cm^{-1}

```
[2]: # Raw spectrum overview
import os
fig, ax = plt.subplots(figsize=(11, 3.5))
ax.plot(x, y, lw=0.7)
ax.set(xlabel="Raman shift [cm-1]", ylabel="Intensity [cps]",
       title=f"{os.path.basename(INFILE)} - full SWNT spectrum "
             f"(note RBM ~200, Si ~520, D, G, 2D)")
```

```
ax.minorticks_on()
plt.show()
```



1.1 Configure and run the pipeline

Adjust `laser_nm` (532/633/785), `gminus_lineshape` (lorentzian for semiconducting, bwf for metallic tubes) or any region in `SwntConfig`.

```
[3]: # RBM shoulders this spectrum hides under stronger neighbours (~190, ~200 on
# the rising flank of the 210 band; ~300) are seeded manually - the prominence
# detector only sees local maxima, so buried shoulders need an explicit hint.
cfg = SwntConfig(laser_nm=532, gminus_lineshape="lorentzian",
                 rbm_extra_centers=(190.0, 200.0, 300.0))
res = analyze_swnt(x, y, cfg)
print(summary_text(res))
```

Si calibration: observed 519.62 cm⁻¹, offset -1.08 cm⁻¹ (nominal 520.7)

RBM peaks (diameter, candidate (n,m)):

194.5 cm ⁻¹	d=1.27 nm	-
205.2 cm ⁻¹	d=1.21 nm	(14,2)M, (15,0)M, (10,7)M
211.6 cm ⁻¹	d=1.17 nm	(15,0)M, (10,7)M, (14,2)M
224.4 cm ⁻¹	d=1.11 nm	(11,5)M, (8,8)M, (12,3)M
234.0 cm ⁻¹	d=1.06 nm	(13,1)M, (12,3)M, (8,8)M
244.9 cm ⁻¹	d=1.01 nm	(9,6)M, (13,1)M
257.7 cm ⁻¹	d=0.96 nm	-
268.0 cm ⁻¹	d=0.93 nm	-
297.3 cm ⁻¹	d=0.83 nm	(9,2)S, (10,0)S
335.2 cm ⁻¹	d=0.74 nm	(6,5)S, (9,1)S, (8,3)S

D-G joint fit ($R^2=0.9991$):

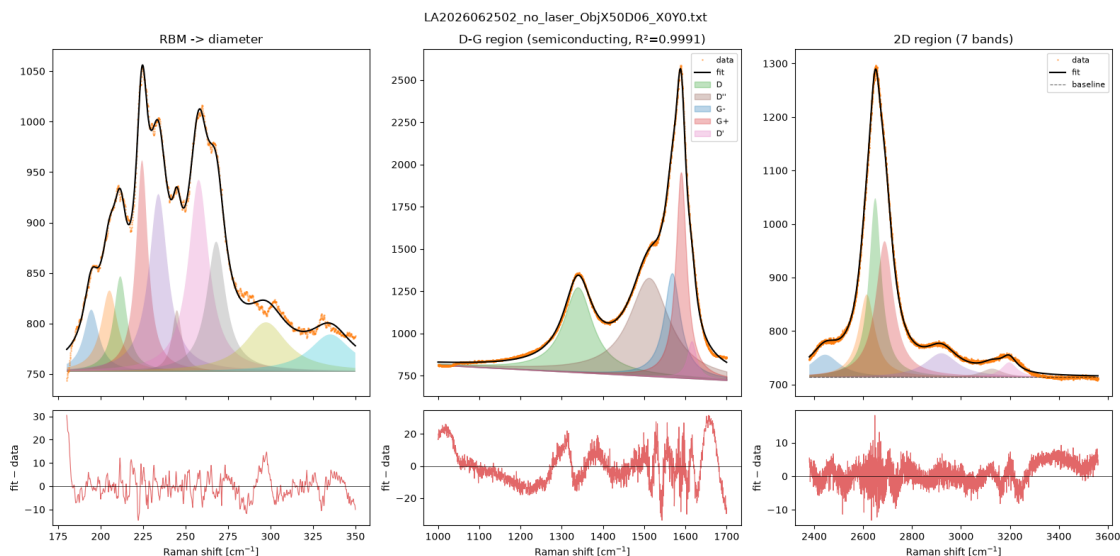
D 1339.2 cm⁻¹ (FWHM 85)
D' 1511.6 cm⁻¹ (FWHM 117, broad disorder band)

G+ 1589.5, G- 1567.7 (split 21.8 cm⁻¹) -> semiconducting
D' 1615.3 cm⁻¹ (FWHM 25)
I(D)/I(G+) = 0.414

2D region (7 components, R²=0.9994):

G* 2444.5 cm⁻¹ (FWHM 144, height 42)
2D 2614.8 cm⁻¹ (FWHM 79, height 155)
2D 2649.4 cm⁻¹ (FWHM 61, height 335)
2D 2687.8 cm⁻¹ (FWHM 88, height 254)
D+D' 2919.8 cm⁻¹ (FWHM 162, height 45)
2D' 3127.1 cm⁻¹ (FWHM 109, height 16)
2D' 3197.1 cm⁻¹ (FWHM 77, height 28)
dominant 2D 2649.4 cm⁻¹, I(2D)/I(G+) = 0.275

```
[4]: # Six-panel decomposition: RBM / D-G / 2D, each with data+fit over its residual
fig = plot_analysis(res, suptitle=os.path.basename(INFILE))
plt.show()
```



1.2 RBM → diameter → candidate (n,m)

(n,m) candidates come from a first-order tight-binding Kataura model and are **approximate** — see `ramanfit_swnt/kataura.py`.

```
[5]: for pk in res["rbm"]["peaks"]:
    nm = ", ".join(f"({c['n']},{c['m']}){c['type']}"
                  for c in pk.get("nm_candidates", [])[:3]) or "-"
    print(f"RBM {pk['center']:7.1f} cm-1 d = {pk['diameter_nm']:.3f} nm □
    ↳{nm}")
```

RBM	194.5 cm ⁻¹	d = 1.275 nm	-
RBM	205.2 cm ⁻¹	d = 1.209 nm	(14,2)M, (15,0)M, (10,7)M
RBM	211.6 cm ⁻¹	d = 1.172 nm	(15,0)M, (10,7)M, (14,2)M
RBM	224.4 cm ⁻¹	d = 1.105 nm	(11,5)M, (8,8)M, (12,3)M
RBM	234.0 cm ⁻¹	d = 1.060 nm	(13,1)M, (12,3)M, (8,8)M
RBM	244.9 cm ⁻¹	d = 1.013 nm	(9,6)M, (13,1)M
RBM	257.7 cm ⁻¹	d = 0.962 nm	-
RBM	268.0 cm ⁻¹	d = 0.925 nm	-
RBM	297.3 cm ⁻¹	d = 0.834 nm	(9,2)S, (10,0)S
RBM	335.2 cm ⁻¹	d = 0.740 nm	(6,5)S, (9,1)S, (8,3)S

1.3 D-G band peaks (joint fit)

Centers, widths and uncertainties for the five components fit jointly over 1000–1700 cm⁻¹: **D**, a broad **D** disorder band filling the D-to-G valley, the **G**/**G** doublet, and **D** (~1600, defect-activated, just above G). Finite **center** ± values confirm the fit is well-determined (not railed).

```
[6]: dg = res["dg"]
print(f"D-G joint fit R² = {dg['r_squared']:.4f} ({'metallic' if
    ↪dg['metallic'] else 'semiconducting'})\n")
print(f"{'peak':4s} {'center':>9s} {'± (cm⁻¹)':>9s} {'FWHM':>7s} {'height':
    ↪>9s}")
for rec in (dg['D'], dg['Dpp'], dg['G_minus'], dg['G_plus'], dg['Dprime']):
    ce = rec.get('center_stderr')
    ce_s = f"{ce:.2f}" if ce is not None else " n/a"
    print(f"{'rec['label']':4s} {'rec['center']':9.1f} {'ce_s:>9s} "
        f"{'rec.get('fwhm', float('nan'))':7.1f} {'rec['height']':9.0f}")
```

D-G joint fit R² = 0.9991 (semiconducting)

peak	center	± (cm ⁻¹)	FWHM	height
D	1339.2	0.07	85.4	503
D''	1511.6	0.34	117.2	581
G-	1567.7	0.33	47.4	616
G+	1589.5	0.06	31.1	1216
D'	1615.3	0.17	25.5	219

1.4 2D region (multi-component)

The whole second-order region (2380–3560 cm⁻¹) is fit with a **flat (constant) baseline** at the true continuum floor — the far plateau past 2D' (~3400–3500 cm⁻¹). The apparent “hump” between the 2D and 2D' bands is not a curved background but real **D+D** (~2915) and **2D** (~3105) combination bands, so once those are fit explicitly the baseline stays horizontal. The band set matches a reference Igor multipeak fit of this sample: **G*** (~2446, iTOLA), the **2D** triplet (~2655, main + two shoulders from the diameter distribution / different (n,m) species), **D+D**, **2D**, and **2D'** (~3185, overtone of D'). The I(2D)/I(G) metric uses the dominant 2D peak.

```
[7]: t2d = res["twod_band"]
print(f"2D region: {len(t2d['peaks'])} components (R2 = {t2d['r_squared']:.4f})\n")
print(f"{'band':5s} {'center':>9s} {'FWHM':>7s} {'height':>9s}")
for pk in sorted(t2d["peaks"], key=lambda r: r["center"]):
    print(f"{pk['label']:5s} {pk['center']:9.1f} {pk['fwhm']:7.1f} {pk['height']:9.0f}")
print(f"\ndominant 2D {t2d['peak']['center']:.1f} cm-1, I(2D)/I(G) = {res['quality']['I2D_IG']:.3f}")
```

2D region: 7 components (R² = 0.9994)

band	center	FWHM	height
G*	2444.5	143.8	42
2D	2614.8	79.3	155
2D	2649.4	61.1	335
2D	2687.8	87.9	254
D+D''	2919.8	161.6	45
2D''	3127.1	109.1	16
2D'	3197.1	77.3	28

dominant 2D 2649.4 cm⁻¹, I(2D)/I(G) = 0.275

1.5 Si calibration & quality metrics

```
[8]: cal = res["calibration"]
g = res["gband"]
q = res["quality"]
print(f"Si calibration applied: {cal['applied']} (offset {cal['offset']:+.2f} cm-1)")
print(f"G = {g['G_plus']['center']:.1f}, G = {g['G_minus']['center']:.1f} cm-1")
print(f"character: {'metallic' if g['metallic'] else 'semiconducting'}")
quality = "low defects" if q['ID_IG'] < 0.2 else ("moderate defects" if q['ID_IG'] < 0.5 else "high defects")
print(f"I(D)/I(G) = {q['ID_IG']:.3f} ({quality})")
print(f"2D = {q['twod_center']:.1f} cm-1, I(2D)/I(G) = {q['I2D_IG']:.3f}")
```

Si calibration applied: True (offset -1.08 cm⁻¹)

G = 1589.5, G = 1567.7 cm⁻¹

character: semiconducting

I(D)/I(G) = 0.414 (moderate defects)

2D = 2649.4 cm⁻¹, I(2D)/I(G) = 0.275

1.6 Save a summary

```
[9]: with open("swnt_summary.txt", "w") as f:
      f.write(summary_text(res))
      print("wrote swnt_summary.txt")
```

wrote swnt_summary.txt